

Quiz # 3

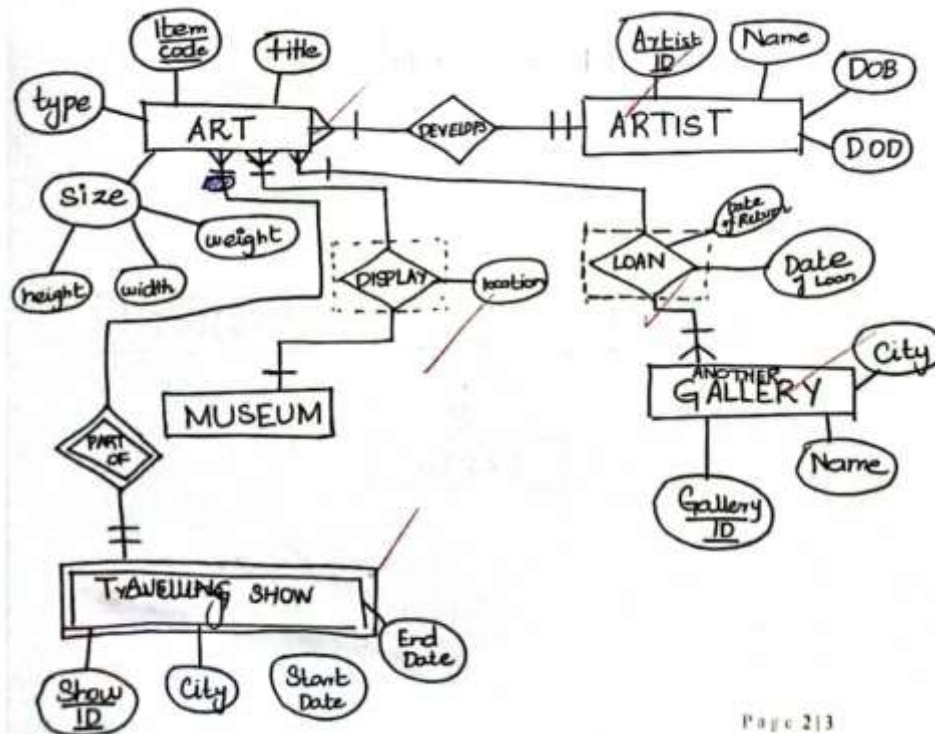
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Content	Clo's	Activity
- Introduction to ERD, Basic concept entity, attributes, relationship - Constraints on relationship, Case studies	2	Quiz 3

Q1: Case Study

An art museum owns a large volume of works of art. Each work of art is described by an item code (identifier), title, type, and size; size is further composed of height, width, and weight. A work of art is developed by an artist, but the artist for some works is unknown. An artist is described by an artist ID (identifier), name, date of birth, and date of death (which is null for still living artists). Only data about artists for works currently owned by the museum are kept in the database. At any point in time, a work of art is either on display at the museum, held in storage, away from the museum as part of a traveling show, or on loan to another gallery. If on display at the museum, a work of art is also described by its location within the museum. A traveling show is described by a show ID (identifier), the city in which the show is currently appearing, and the start and end dates of the show. Many of the museum works may be part of a given show, and only active shows with at least one museum work of art need be represented in the database. Finally, another gallery is described by a gallery ID (identifier), name, and city. The museum wants to retain a complete history of loaning a work of art to other galleries, and each time a work is loaned, the museum wants to know the date the work was loaned and the date it was returned. As you develop the ERD for this problem, follow good data naming guidelines.





Q2: Case Study

The Country Bus Company owns several buses. The attributes of the bus are Registration number, Engine number, vehicle id, no of seats, capacity and single or double desk. Each bus is allocated to a particular route, although some routes may have several buses. The route has the following characteristics route id, name and average number of passengers carried on each route. Each route passes through several towns. Towns have id, name attributes. One or more drivers are allocated to each stage of the route, which corresponds to the journey through some or all the towns on a route. Drivers has the following attributes (driver id, name, address, contact number, driver license number, address experience). Some of the towns have garages, where buses are kept. Garage has the following attributes id, name and location, capacity. Each bus has a schedule for the departure from one location to another. Schedule contain bus id, departure location, arrival location, departure time and arrival time.

Entities

- Bus - Company owns busses and will hold information about them.
- Route - Buses travel on routes and will need described.
- Town - Buses pass through towns and need to know about them
- Driver - Company employs drivers, personnel will hold their data.
- Stage - Routes are made up of stages
- Garage - Garage houses buses, and need to know where they are.

Relationships

- A bus is allocated to a route and a route may have several buses.
- Bus-route (m:1) is serviced by
- A route comprises of one or more stages.
- route-stage (1:m) comprises
- One or more drivers are allocated to each stage.
- driver-stage (m:1) is allocated
- A stage passes through some or all of the towns on a route.
- stage-town (m:n) passes-through
- A route passes through some or all of the towns
- route-town (m:n) passes-through
- Some of the towns have a garage
- garage-town (1:1) is situated
- A garage keeps buses and each bus has one 'home' garage
- garage-bus (m:1) is garaged

Attributes

- Bus (reg-no,make,size,deck,no-pass)
- Route (route-no,avg-pass)
- Driver (emp-no,name,address,tel-no)
- Town (name)
- Stage (stage-no)
- Garage (name,address)

